

7.3 Compression and Deployment

Prune, quantize, distill, then speculative decoding at serving time.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Scaling (note 7.1) makes a model that may not fit on the box you have. **Efficiency** is how you ship it: fewer weights, fatter arithmetic, a smaller student, or fewer expensive forward passes at decode time.

1. Shrink the weights

Prune: set small weights to zero (unstructured) or drop heads, neurons, or layers (structured). You need a pass of recovery training or the quality drop is ugly.

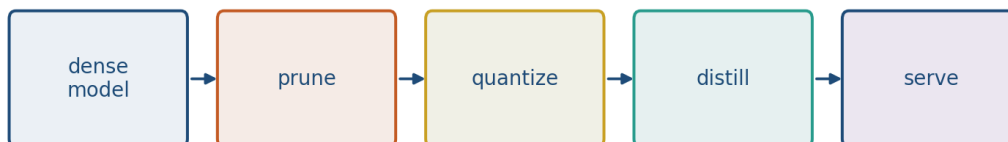
Quantize: store weights (and sometimes activations) in fewer bits. INT8 and 4-bit GPTQ/AWQ are the ones you will see in Hugging Face. Affine quantization uses a **scale** and **zero-point**:

$$x \approx s(q - z), \quad q \in \{0, \dots, 2^b - 1\}.$$

Lab 7 does this on a toy tensor and reports MSE. The curve is not free: outliers in activations break naive INT8.

Distill: a frozen **teacher** produces logits or hidden states; a smaller **student** matches them (plus the usual labels). You buy a model that is cheaper to run, not a new capability.

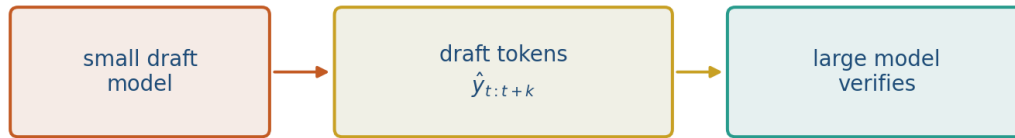
Compression is a pipeline. Measure quality after each cut.



2. Speculative decoding

Autoregressive decode is serial: one token, then another. **Speculative decoding** runs a cheap **draft** model for several tokens, then the large **target** checks them in one parallel pass. If the target agrees, you accept a block; if not, you roll back at the first mismatch and sample from the target. Expected speedup is $> 1 \times$ when the draft is often right; the **distribution** is still the target's.

Speculative decoding: guess many tokens, accept the prefix the big model agrees on



This is not pruning. It is extra compute on a small net to save steps on a large net.

3. What to report

Name bits, which layers you quantized, whether you distilled, and tokens per second versus a dense fp16 baseline. Quality: the same eval as the uncompressed model, not only perplexity. For a project, a 4-bit laptop demo is a deployment result; it is not a new architecture.

4. Practice

1. Why does a scale-and-zero-point map need z , not only s , if x is not centered at zero?
2. Distillation needs a teacher. What happens if the teacher is already calibrated badly on your domain?
3. Speculative decoding must not change the target distribution. Where does a rejected draft token get replaced?